

CONFORMAL UNCERTAINTY FOR NO-REFERENCE MOS PREDICTION UNDER DISTRIBUTION SHIFT: A LEAKAGE-CONTROLLED STUDY

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ABSTRACT

No-reference mean opinion score (MOS) predictors output a single number with no signal of trustworthiness, a risk in automated pipelines. We test whether distance from a reference feature distribution can supply that signal, using three frozen predictors, DNSMOS P.835, NISQA, and UTMOS, in a leakage-controlled design that separates feature-reference fitting, content-disjoint calibration, and evaluation across all seven NISQA Corpus subsets. Euclidean, Mahalanobis, and nearest-neighbor distances detect which corpus a clip came from (mean domain AUROC 0.615–0.690) but do not flag which clips carry the largest prediction error within a corpus (mean AUROC 0.433–0.499, near chance). Split conformal intervals built on these distances reach 90.1–91.8% coverage at the 90% nominal level with widths of 2.26–2.91 MOS on the four-point scale. Stratifying interval width by Mahalanobis distance helps some predictor–domain pairs and hurts others. This pattern reproduces on an independent multilingual speech-enhancement corpus, so practitioners should read a feature-distance alarm as a domain-mismatch flag, not a per-utterance reliability score.

Index Terms— mean opinion score, conformal prediction, out-of-distribution detection, distribution shift, uncertainty quantification

1. INTRODUCTION

No-reference mean opinion score (MOS) predictors are increasingly used to screen and compare speech systems at scale. DNSMOS P.835 [1], NISQA [2], and UTMOS [3] reduce reliance on repeated large-scale listening tests, while DNSMOS also serves as an official metric in the DNS Challenge [4, 5]. Their use has outpaced scrutiny of their failure modes.

Existing MOS predictors return point estimates without an explicit reliability measure, so a practitioner cannot tell from the score alone whether an utterance should be trusted, reviewed, or sent to a listening test. Feature-space distance is an attractive post-hoc signal because it requires neither predictor retraining nor labeled shifted data, but its interpretation is not automatic: a score that recognizes a change in language, channel, or degradation process may still fail to identify which individual MOS predictions are inaccurate.

We test this hypothesis through a conformal uncertainty wrapper for frozen MOS predictors. Split conformal prediction [6] provides finite-sample marginal coverage when calibration and test examples are exchangeable and calibration labels are independent of model fitting: the wrapper first constructs a global interval from calibration residuals, then applies an established Mondrian-style construction [7] in which Mahalanobis distance [8] selects a distance-specific interval radius. The central question is whether a feature-space domain-shift signal transfers to utterance-level error ranking strongly enough to support this adaptation.

This work makes three contributions. We build a leakage-controlled evaluation with disjoint feature-reference, calibration, and in-domain evaluation roles. We compare Euclidean, Mahalanobis, and nearest-neighbor scores across domain detection and utterance-level error detection for three MOS predictors. We then test the practical objective directly by comparing fixed and distance-stratified conformal intervals under five metadata-defined shifts. Mean domain-detection AUROC is 0.615–0.690, while mean AUROC for detecting the highest-error quartile is only 0.433–0.499, and interval adaptation is predictor and domain dependent. The contribution is therefore both methodological and empirical: it shows why an out-of-distribution (OOD) score must be validated as a signal of prediction error, not merely of domain membership, before it is used to adapt prediction uncertainty. Code is available at <https://github.com/fransfela/mos-conformal>.

2. RELATED WORK

Early no-reference speech quality estimators modeled distortion type and signal statistics [9]. NISQA [2] uses self-attention trained on multi-condition MOS data, DNSMOS P.835 [1] predicts speech, background, and overall quality by extending the earlier single-score DNSMOS metric [10], and UTMOS [3] builds on self-supervised representations.

OOD detection includes confidence-based [11], density-based, and distance-based methods. Lee et al. [8] use class-conditional Mahalanobis distance for image classifiers, and deep nearest-neighbor distance has been evaluated for audio OOD detection [12], following the deep k -NN formulation [13]. We study a regression analogue based on one Gaus-

sian, compared with Euclidean and nearest-neighbor scores without retraining the predictor, and separately test whether domain discrimination transfers to prediction-error ranking.

Conformal prediction converts residual scores into intervals with finite-sample marginal coverage under exchangeability [14, 6, 15]. Mondrian conformal regressors bin a difficulty estimate and calibrate a radius within each bin [7]. Weighted conformal prediction restores a coverage guarantee under a known covariate shift [16], but requires an importance-weight estimate that we do not attempt here. Recent MOS work predicts aleatoric and epistemic uncertainty for selective prediction and domain detection [17], or learns a predictive MOS distribution and filters uncertain samples [18]. We do not present distance binning, audio OOD detection, or MOS selective prediction as new methods. The novelty lies in testing, without retraining or shifted labels, whether internal feature distance transfers across domain detection, utterance-level error ranking, and conformal interval adaptation in frozen speech-quality predictors.

3. METHOD

For a speech waveform $\mathbf{x}$, a frozen no-reference predictor f returns the scalar MOS estimate $\hat{q} = f(\mathbf{x})$. We extract a fixed-dimensional representation $\mathbf{z} = \phi(\mathbf{x}) \in \mathbb{R}^d$ from each predictor’s penultimate layer, mean-pooling frame- or segment-level activations: the second dense layer before the final head for DNSMOS ($d = 64$), the self-attention time-dependency module before the pooling heads for NISQA ($d = 64$), and the BiLSTM output of the LDConditioner for UTMOS ($d = 1024$). Each predictor has its own feature space and therefore its own reference distribution.

3.1. Feature-Space Scores

For each predictor, T-SIM supplies reference features $\{\mathbf{z}_i\}_{i=1}^N$, where $N = 10,000$. The maximum-likelihood empirical mean and covariance are

$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i, \quad (1)$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{z}_i - \hat{\boldsymbol{\mu}})(\mathbf{z}_i - \hat{\boldsymbol{\mu}})^\top. \quad (2)$$

Here $N > d$ for all predictors, with $d = 64$ for DNSMOS and NISQA ($N/d = 156$) and $d = 1024$ for UTMOS ($N/d = 9.8$). We use the empirical covariance throughout and do not apply Ledoit-Wolf shrinkage [19]. At UTMOS’s sample size, shrinkage changes AUROC by at most 0.09 (domain) and 0.12 (error), both above the 0.075 spread across scores in Table 3, so per-predictor rankings are not resolvable here. The near-chance conclusion of Section 4.3 is unchanged under either estimator. The Mahalanobis score is

$$d_M(\mathbf{x}) = \left[(\mathbf{z} - \hat{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{z} - \hat{\boldsymbol{\mu}}) \right]^{1/2}. \quad (3)$$

We also evaluate Euclidean distance $\|\mathbf{z} - \hat{\boldsymbol{\mu}}\|_2$ and the distance to the nearest T-SIM representation [13], fixing $k = 1$. A repository sweep over $k \in \{1, \dots, 50\}$ changes mean AUROC by at most 0.004. A larger score is prespecified to mean farther from the reference distribution. These scores quantify representation shift rather than estimate MOS error.

3.2. Split Conformal Interval

V-SIM is partitioned by content identifier into disjoint calibration and in-domain evaluation halves of 1,250 clips each. Let $\mathcal{D}_{\text{cal}} = \{(\mathbf{x}_j, q_j^*)\}_{j=1}^m$ denote the calibration half, where $m = 1,250$ and q_j^* is subjective MOS. The absolute-residual nonconformity score is

$$s_j = |q_j^* - \hat{q}_j|, \quad \hat{q}_j = f(\mathbf{x}_j). \quad (4)$$

Let $s_{(1)} \leq \dots \leq s_{(m)}$ be the ordered calibration scores and let $\alpha \in (0, 1)$ denote the miscoverage rate, so nominal coverage is $1 - \alpha$ (we use $\alpha = 0.10$). For $\alpha \geq 1/(m + 1)$, define

$$k_\alpha = \lceil (m + 1)(1 - \alpha) \rceil, \quad \hat{Q}_{1-\alpha} = s_{(k_\alpha)}. \quad (5)$$

The fixed-width interval is

$$C_\alpha(\mathbf{x}) = [\hat{q} - \hat{Q}_{1-\alpha}, \hat{q} + \hat{Q}_{1-\alpha}]. \quad (6)$$

We do not clip the endpoints to $[1, 5]$, since clipping can alter coverage at the scale boundaries. If the calibration examples and a new example are exchangeable and calibration labels were not used to fit or select f , then split conformal prediction gives the marginal guarantee

$$\Pr\{q^* \in C_\alpha(\mathbf{x})\} \geq 1 - \alpha. \quad (7)$$

The content-disjoint split prevents direct content leakage but does not itself prove exchangeability, which can in principle be tested online [20], though we do not conduct that test here. The NISQA model was developed using NISQA Corpus partitions [2], so its intervals are treated as empirical rather than as a fresh-data guarantee.

3.3. Distance-Stratified Interval

Following Mondrian conformal regression with binned difficulty estimates [7], we test whether Mahalanobis distance supports instance-dependent widths. We select Mahalanobis distance because Section 4.3 shows it is comparable to nearest-neighbor distance on domain-detection AUROC while needing only a fixed (d, d) covariance matrix rather than a reference bank that grows with the calibration set. The calibration examples are divided into K empirical quantile bins of $d_M(\mathbf{x})$. Let I_k be the calibration indices assigned to bin k , let $m_k = |I_k|$, and let $s_{(r)}^{(k)}$ be the r th ordered residual within that bin. For $\alpha \geq 1/(m_k + 1)$,

$$\ell_{\alpha,k} = \lceil (m_k + 1)(1 - \alpha) \rceil, \quad \hat{Q}_{1-\alpha}^{(k)} = s_{(\ell_{\alpha,k})}^{(k)}. \quad (8)$$

Table 1. NISQA Corpus subsets [2]. V-SIM splits deterministically into disjoint calibration and in-domain evaluation halves. N : clips with MOS labels.

Abbrev.	Subset	Role	N
T-SIM	TRAIN_SIM	Feature reference	10 000
T-LIV	TRAIN_LIVE	Shifted evaluation	1 020
V-SIM-Cal	VAL_SIM	Calibration	1 250
V-SIM-ID	VAL_SIM	ID evaluation	1 250
V-LIV	VAL_LIVE	Shifted evaluation	200
T-FOR	TEST_FOR	Shifted evaluation	240
T-TLK	TEST_LIVETALK	Shifted evaluation	232
T-P501	TEST_P501	Shifted evaluation	240

If $b(\mathbf{x}) \in \{1, \dots, K\}$ is the bin assigned using the calibration-derived boundaries, then

$$C_{\alpha}^{\text{adapt}}(\mathbf{x}) = [\hat{q} - \hat{Q}_{1-\alpha}^{(b(\mathbf{x}))}, \hat{q} + \hat{Q}_{1-\alpha}^{(b(\mathbf{x}))}]. \quad (9)$$

Scores outside the calibration range enter the first or last bin. We use $K = 5$ and hold the boundaries and radii fixed at evaluation. Because the boundaries are estimated from the calibration covariates and shifted test examples need not be exchangeable with calibration examples, we treat Eq. (9) as a diagnostic post-hoc construction. We do not claim conditional or arbitrary-shift coverage for it.

4. EXPERIMENTS

4.1. Setup

We use all seven subsets of the NISQA Corpus [2], summarized in Table 1. T-SIM fits the feature reference statistics and supplies the k -NN reference bank. A deterministic content-disjoint split assigns 1,250 V-SIM clips to conformal calibration and 1,250 to in-domain evaluation. T-LIV, V-LIV, T-FOR, T-TLK, and T-P501 are evaluated as metadata-defined shifts in recording condition, language, or channel. No label from a shifted subset is used for calibration.

We evaluate DNSMOS P.835 [1] using its overall-quality output, NISQA v2 [2], and UTMOS [3]. All outputs are represented on the 1–5 MOS scale. DNSMOS predictions include the non-personalized P.835 polynomial mapping.¹ UTMOS scores come from a torchaudio reimplementation of the released checkpoint rather than the pinned fairseq scorer, and we have not verified numerical equivalence between the two.

For point predictions, we report SRCC and RMSE. Domain detection compares each shifted subset with held-out V-SIM using AUROC and FPR at 95% TPR. V-SIM has no domain-detection bar because it defines the in-domain negative class. Error detection is a separate within-domain task. In each evaluation subset, positives are the highest absolute-error quartile and the score direction is fixed in advance so that larger distance predicts larger error. Figure 1 reports group-bootstrap

¹The stored clip-level correction approximates segment-wise mapping for multi-segment clips, with mean absolute discrepancy 0.0038 MOS in a stratified verification sample.

Table 2. Point-prediction performance (higher SRCC, lower RMSE better). **Bold:** best per column.

Model	Metric	Held-out ID	Shifted evaluation				
		V-SIM	T-LIV	V-LIV	T-FOR	T-TLK	T-P501
DNSMOS	SRCC	0.703	0.674	0.674	0.541	0.593	0.695
	RMSE	0.854	0.549	0.525	0.845	0.765	0.913
NISQA	SRCC	0.805	0.649	0.638	0.725	0.706	0.831
	RMSE	0.860	0.528	0.525	0.734	0.747	0.653
UTMOS	SRCC	0.830	0.702	0.749	0.769	0.773	0.880
	RMSE	0.662	0.829	0.852	0.604	1.022	0.636

Table 3. Domain detection averaged over five shifted subsets vs. held-out V-SIM (higher AUROC, lower FPR@95 better). **Bold:** best per column.

OOD score	AUROC $\uparrow$				FPR@95 $\downarrow$
	DNS	NIQ	UTM	Avg.	Avg.
L2 Euclidean	0.585	0.693	0.569	0.615	0.854
Mahalanobis	0.668	0.803	0.561	0.678	0.732
k -NN ($k = 1$)	0.697	0.805	0.569	0.690	0.763

95% confidence intervals from 2,000 resamples. Interval comparisons independently resample calibration and evaluation groups, then refit quantiles, bin boundaries, and bin-specific radii in every bootstrap replicate.

4.2. Point Prediction

Table 2 shows that a metadata-defined shift does not imply uniform degradation, though NISQA was trained on partitions of this corpus, so its numbers are an in-sample reference rather than a generalization test comparable to DNSMOS and UTMOS. UTMOS has the highest SRCC on every subset but reaches RMSE 1.022 on T-TLK. DNSMOS attains nearly the same V-LIV RMSE as NISQA, yet has substantially higher error on T-FOR and T-P501.

4.3. Domain Detection Versus Error Detection

Table 3 evaluates domain discrimination. Averaged over 15 predictor–shift pairs, AUROC is 0.615 for Euclidean distance, 0.678 for Mahalanobis distance, and 0.690 for nearest-neighbor distance. NISQA is in-sample (Section 4.2). Excluding it, mean Mahalanobis AUROC over DNSMOS and UTMOS alone is 0.614, not 0.678. Nearest-neighbor distance attains the highest per-predictor AUROC in all three columns, though the margin over Mahalanobis is small for NISQA and negligible for UTMOS, while Mahalanobis obtains the lowest mean FPR@95. Neither score dominates both metrics, so we treat Mahalanobis and nearest-neighbor distance as comparable alternatives for domain detection.

Figure 1 shows that domain discrimination and error detection are not interchangeable. Across the 18 predictor–subset pairs, mean error-detection AUROC is 0.478 for Mahalanobis distance. The corresponding means are 0.433 for Euclidean

Table 4. Coverage (%) at 90% nominal level. Fixed (F) and five-bin Mahalanobis-adaptive (A) intervals share the V-SIM calibration split. Last row: held-out V-SIM coverage at 80% nominal. Widths reported in text (Section 4.4).

Dataset	DNSMOS		NISQA		UTMOS	
	F	A	F	A	F	A
V-SIM	90.2	90.1	90.1	89.4	91.8	92.1
T-LIV	99.0	98.6	99.3	94.3	81.3	82.1
V-LIV	100.0	99.5	99.5	98.0	78.0	79.0
T-FOR	90.0	87.9	92.9	90.8	92.1	93.3
T-TLK	94.4	93.5	93.5	84.1	66.4	67.7
T-P501	85.8	87.1	97.1	90.4	92.1	91.2
V-SIM (80%)	80.2	80.1	79.8	81.0	79.8	80.2

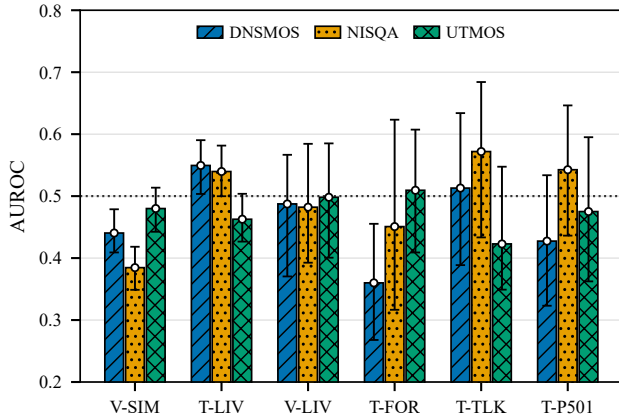


Fig. 1. Within-domain detection of highest absolute-error quartile, Mahalanobis distance (Euclidean, nearest-neighbor, extremity in text). Color/hatch: predictor. Points: AUROC. Error bars: group-bootstrap 95% CI, 2,000 resamples. V-SIM included for error detection, not domain discrimination.

distance, 0.499 for nearest-neighbor distance, and 0.421 for prediction extremity $|\hat{q} - 3|$. On T-LIV, Mahalanobis AUROC is 0.549 for DNSMOS, 0.540 for NISQA, and 0.463 for UTMOS. DNSMOS on T-FOR reaches only 0.360 with 95% CI [0.268, 0.455], and NISQA on its own held-out V-SIM half reaches 0.385 with CI [0.349, 0.418], an equally sub-chance and even tighter interval on the model’s in-sample data. With score direction prespecified before evaluation, both intervals exclude 0.5, so this is a sub-chance finding rather than a null result: the scores are systematically anti-correlated with error on these subsets. Absolute error is largest for mid-scale predictions (extremity AUROC 0.421), so a score that tracks typicality rather than extremity is anti-correlated with error by construction. Spearman correlation between $d_M(\mathbf{x})$ and $|\hat{q} - q^*|$ within each subset, a continuous rank-correlation check, agrees qualitatively with the AUROC-based conclusion: correlations are small and inconsistent in sign across predictors and subsets. A single higher-distance-means-higher-error rejection rule is therefore not transferable.

Applying the NISQA-Corpus-fitted Mahalanobis, Eu-

clidean, and nearest-neighbor references, without refitting, to 364 clips from the Interspeech 2025 URGENT Challenge [21] gives domain-detection AUROC up to 0.911 against held-out V-SIM but error-detection AUROC of only 0.415 to 0.541. This dissociation is not specific to NISQA Corpus.

4.4. Conformal Intervals

Table 4 reports the fixed and five-bin Mahalanobis-adaptive intervals. Fixed coverage on held-out V-SIM is 90.2%, 90.1%, and 91.8% for DNSMOS, NISQA, and UTMOS. This confirms the expected in-domain behavior of the disjoint calibration design. Fixed widths span 2.26 to 2.91 MOS, 56–73% of the four-point (1–5) scale, so the coverage guarantee is bought with intervals that occupy most of the usable range. Fixed width is constant per predictor by construction. Adaptive width instead varies by subset, most for NISQA (2.14–2.91 MOS). At 80% nominal, held-out V-SIM fixed coverage is 80.2%, 79.8%, and 79.8% for DNSMOS, NISQA, and UTMOS, with width shrinking to 2.30, 2.30, and 1.69 MOS. Coverage under shift ranges from 66.4% to 100.0%, which is consistent with the absence of an arbitrary-shift guarantee.

Adaptation is predictor- and domain-dependent. For NISQA, coverage changes from 97.1% to 90.4% on T-P501 and from 93.5% to 84.1% on T-TLK, with both bootstrap CIs excluding zero. Although k -NN and Euclidean bins avoid the T-TLK undercoverage, the results do not identify a superior wrapper. We therefore evaluate ensemble disagreement as a stronger unsupervised difficulty score. Computed as the mean absolute leave-one-out deviation across the three frozen predictions, it requires no new inference. Mean error-detection AUROC is 0.626 over the same 18 predictor–subset pairs, above 0.478 for Mahalanobis distance, though it falls below chance for NISQA on T-TLK (0.420). Stratifying Mondrian bins by ensemble disagreement raises NISQA T-TLK coverage from 84.1% to 92.2%, but does not uniformly outperform Mahalanobis-adaptive intervals elsewhere: a stronger average error-detection signal, not a reliably transferable one.

5. CONCLUSION

Feature-space distance separates some corpus shifts, but its link to prediction error is weak and can reverse, so Mahalanobis-stratified intervals help some predictor–domain pairs and harm others. This dissociation also holds on an independent corpus. Fixed split conformal intervals give matched-domain coverage under exchangeability, but arbitrary shift has no coverage guarantee. Beyond the UTMOS scorer-fidelity gap noted in Section 4.1, the wrapper assumes calibration-test exchangeability that we do not test here and bins on a fixed Gaussian distance rather than a learned difficulty score. Testing exchangeability online and replacing Mahalanobis distance with a learned, calibrated difficulty estimator are natural next steps.

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